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A Survey on Structured Sparse Attention Methods in Transformers

Transformer models are commonly used in machine learning for tasks such as natural language processing, computer vision, and other areas that involve sequential data. While they are highly effective, their attention mechanism can become very slow and consume a large amount of memory when working with long sequences. This can make it challenging to process lengthy documents, extended conversations, or large datasets efficiently.

Sparse Transformers simplify attention by letting each part of the input focus on only a few parts, instead of all of them. These selected connections are chosen so the model can still understand the important relationships in the data. This saves time and memory while keeping the results accurate. Structured sparse attention mechanisms take this idea a step further, instead of letting every part of the input interact with all other parts, they use predefined patterns such as blocks, sliding windows, or global tokens to guide which parts of the input interact with each other. By following these structured patterns the model can efficiently preserve important context and reduce memory, enabling efficient processing of long sequences.

This survey will focus on Structured Sparse Attention Methods in Transformer Architectures. It will examine early research on structured sparse attention as well as more recent studies that develop or apply these methods in practice. The analysis will categorize approaches based on the type of structured sparsity and evaluate how efficiently each method operates. It will also consider how different designs balance memory and accuracy.

Although structured sparse attention improves efficiency, it can make it harder for the model to capture the full context of the input. In this survey we want to understand, how different structured sparse attention patterns enable transformer models to handle long sequences effectively without compromising important information. This study will work to summarize current research, compare different methods, and outline challenges as well as potential areas for future work.